

Rajabharana Jewellery System — Chen Notation ER Diagram

Database mapping · 15 entities · Full attributes · Billing + Reports

□ Entity (Rectangle) ○ Attribute (Ellipse) ◆ Relationship (Diamond) ■ Implemented ■ Planned Billing/Payment
■ Reports

Figure 1 — Chen Relationship Diagram (Database Tables)

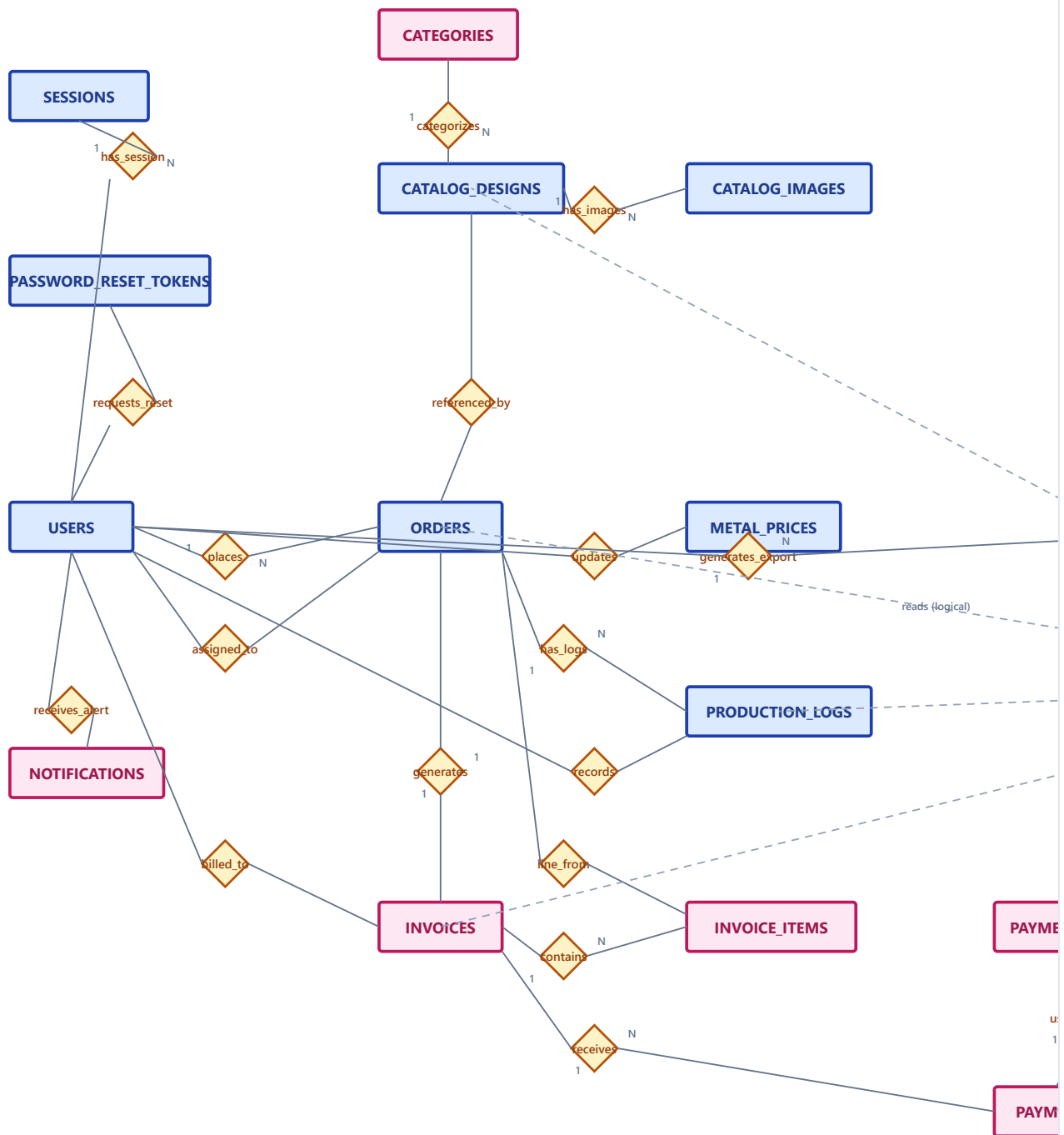


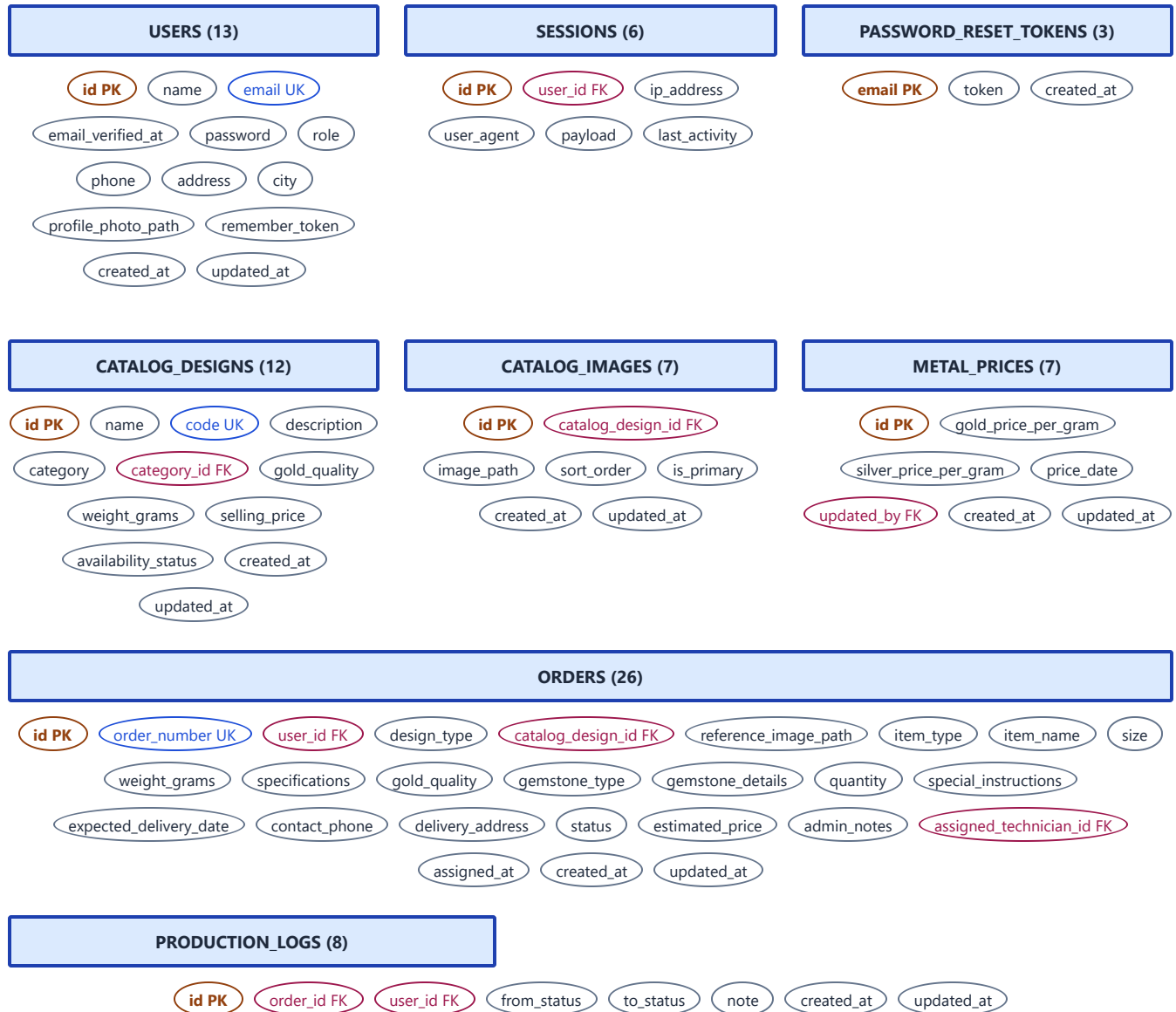
Figure 2 — Entity Attributes (Chen Ellipses) — Auth & Core ✓

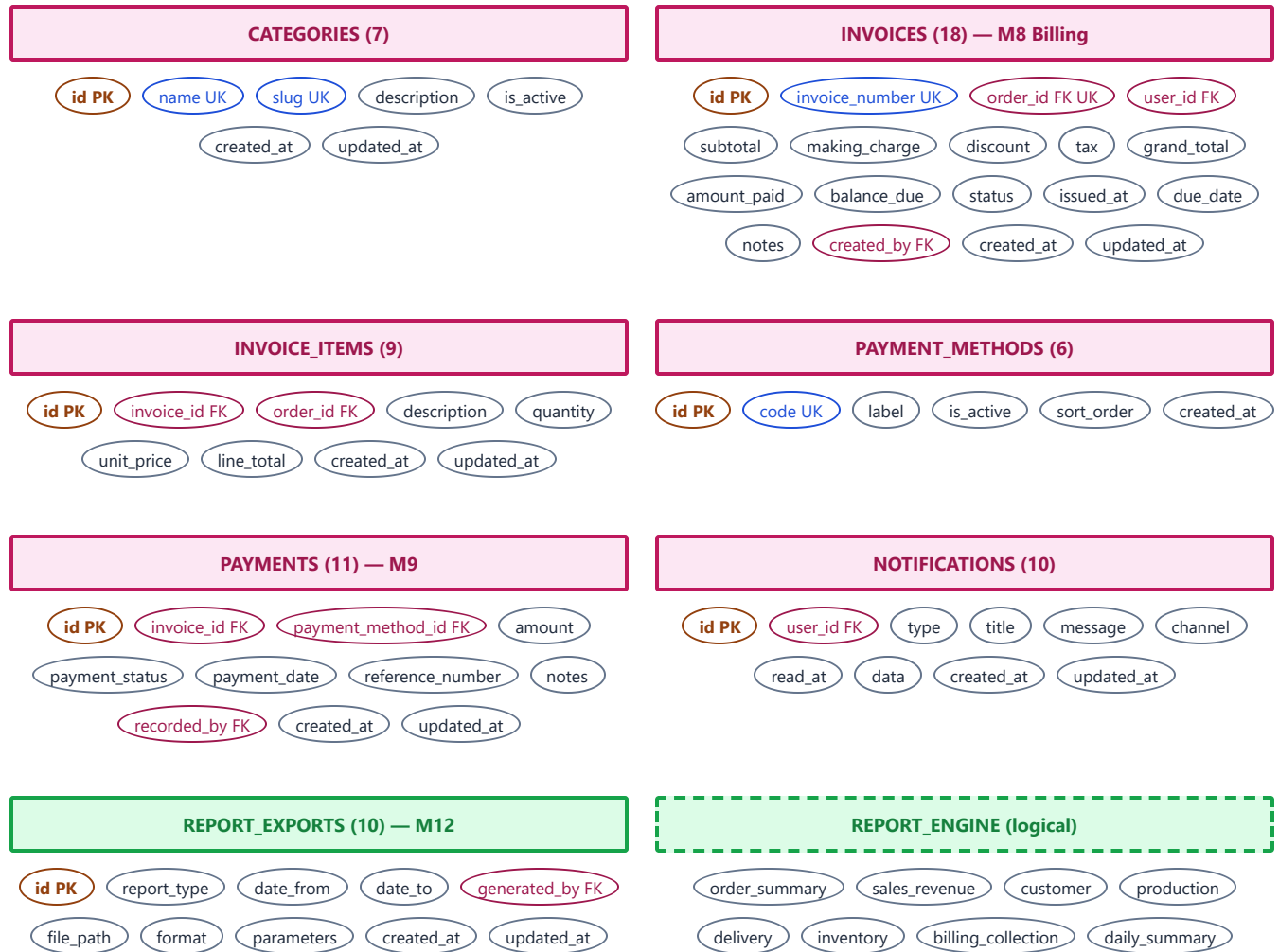
Figure 3 — Entity Attributes — Billing · Payment · Reports 

Figure 4 — Relationships (Diamonds) with Cardinality

Diamond	Entity A	Card.	Entity B	FK	Module
has_session	USERS	1:N	SESSIONS	user_id	Auth
requests_reset	USERS	1:0..1	PASSWORD_RESET_TOKENS	email	Auth
places	USERS	1:N	ORDERS	user_id	Orders
assigned_to	USERS	1:N	ORDERS	assigned_technician_id	Workshop
records	USERS	1:N	PRODUCTION_LOGS	user_id	Workshop
updates	USERS	1:N	METAL_PRICES	updated_by	Metal Price
billed_to	USERS	1:N	INVOICES	user_id	Billing
creates	USERS	1:N	INVOICES	created_by	Billing
records_payment	USERS	1:N	PAYMENTS	recorded_by	Payment
receives_alert	USERS	1:N	NOTIFICATIONS	user_id	Notification
generates_export	USERS	1:N	REPORT_EXPORTS	generated_by	Reports
categorizes	CATEGORIES	1:N	CATALOG_DESIGNS	category_id	Inventory
has_images	CATALOG_DESIGNS	1:N	CATALOG_IMAGES	catalog_design_id	Catalogue
referenced_by	CATALOG_DESIGNS	1:N	ORDERS	catalog_design_id	Orders
has_logs	ORDERS	1:N	PRODUCTION_LOGS	order_id	Workshop
generates	ORDERS	1:1	INVOICES	order_id	Billing
contains	INVOICES	1:N	INVOICE_ITEMS	invoice_id	Billing
line_from	ORDERS	1:N	INVOICE_ITEMS	order_id	Billing
receives	INVOICES	1:N	PAYMENTS	invoice_id	Payment
uses_method	PAYMENT_METHODS	1:N	PAYMENTS	payment_method_id	Payment
reads	REPORT_ENGINE	M:N	ALL TABLES	—	Reports (logical)

Open in browser → Print → PDF for your project report.

Chen notation: Rectangle = Entity (MySQL table) · Ellipse = Attribute · Diamond = Relationship · PK/UK/FK underlined in colour.

Also see: docs/DATABASE_ER_FULL.html (Crow's Foot/Mermaid) · docs/CHEN_ER_DIAGRAM.html (user-role conceptual diagram).

Physical DB: 8 tables implemented · 7 planned (billing, payment, notifications, categories, report_exports).